Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/11/2005
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Functional genomics of plant response and adaptation to low atmospheric pressure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	robferl@ufl.edu	**Fax:**	FY 352-392-4072
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1928
Organization Name:	University of Florida		
PI Address 1:	Horticultural Sciences, 1301 Fifield Hall		
PI Address 2:	Box 110690		
PI Web Page:	http://www.hos.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE to 6/30/2007, per A. Chu 8/8/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNA04CC61G		
Performance Goal No.:			
Performance Goal Text:			
	This proposal seeks a fundamental understanding of the mechanisms involved in plant responses and adaptations to low pressure environments, with the ultimate goal of producing plants that are specifically designed to thrive in low pressures. The essential drivers of this project are 1) the inescapable engineering limitations to producing orbital, lunar or Martian plant growth facilities that contain earth-normal atmospheric pressures, and 2) the knowledge that plants in certain circumstances can live under reduced pressures. In the initial phases of the project, we propose to use the tools of functional genomics in the model plant system of arabidopsis to examine molecular genetic responses that occur during initial and chronic exposure to atmospheres ranging from near earth normal to near Martian pressures and containing various gas mixtures. Based on the results from these analyses, we will identify genes and metabolic pathways that are activated and repressed in response to low-pressure atmospheres, and predict alterations that might enhance the		

Task Description:	adaptation process. Finally, we will select and directly engineer genetic changes that will test these predictions and potentially lead to plants with enhanced abilities to live in low-pressure environments. The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems. There are four specific aims: 1. determine the spectrum of genes activated and repressed by low pressure 2. develop an understanding of the metabolic pathways involved in response and adaptation 3. determine the effects of defined gas mixtures in mitigating the impacts of low pressure atmospheres 4. develop and test biological engineering strategies to enhance growth in low atmospheric pressures The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of plant responses to extreme environments at the molecular genetic level contibutes to basic molecular biology research. This research will also improve our ability to develop more successful closed system ecologies (green houses etc) on earth as we come to a better understanding of how transpiration, carbon fixation and ethylene metabolism are impacted by reduced atmospheric pressures.
Task Progress:	Description and summary of research: This proposal seeks a fundamental understanding of the mechanisms involved in plant responses and adaptations to low pressure environments, with the ultimate goal of producing plants that are specifically designed to thrive in low pressures. The essential drivers of this project are 1) the inescapable engineering limitations to producing orbital, lunar or Martian plant growth facilities that contain earth-normal atmospheric pressures, and 2) the knowledge that plants in certain circumstances can live under reduced pressures. In the initial phases of the project, we propose to use the tools of functional genomics in the model plant system of arabidopsis to examine molecular genetic responses that occur during initial and chronic exposure to atmospheres ranging from near earth normal to near Martian pressures and containing various gas mixtures. Based on the results from these analyses, we will identify genes and metabolic pathways that are activated and repressed in response to low-pressure atmospheres, and predict alterations that might enhance the adaptation process. Finally, we will select and directly engineer genetic changes that will test these predictions and potentially lead to plants with enhanced abilities to live in low-pressure environments. The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems. This research will also improve our ability to develop more successful closed system ecologies (green houses etc) on earth as we come to a better understanding of how transpiration, carbon fixation and ethylene metabolism are impacted by reduced atmospheric pressures. There are four specific aims: 1. determine the spectrum of genes activated and repressed by low pressure 2. develop an understanding of the metabolic pathways involved in response and adaptation 3. determine the effects of defined gas mixtures in mitigating the impacts of low pressure atmospheres 4. develop and test biological engineering strategies to enhance growth in low atmospheric pressures Preliminary experiments set the stage for the design of the current proposal. Experiments utilizing plants of various ages exposed to low pressure and low oxygen environments have begun to fulfill specific aim 1 and address aims 2 and 3. An experiment utilizing 9 day old plants grown vertically on nutrient agar media was designed to evaluate the effects of hypobaric (10 kPa) and hypoxic (2% oxygen) environments on global gene expression patterns. These experiments demonstrate that Arabidopsis plants respond to hypobaria with extensive changes in gene expression patterns, indicating that hypobaria elicits a dynamic adaptive response. Some of the altered gene expression patterns are similar to those involved in the hypoxic stress response, as would be expected from the low oxygen partial pressures present in hypobaria; yet some changes in gene expression patterns are unique to hypobaric conditions. These unique differential gene expression patterns demonstrate that hypobaria is not equivalent to hypoxia as an abiotic stress and suggest that, for plants, the response to hypobaria is more complex than the acclimation to the reduced partial pressures of oxygen inherent to low atmospheric pressures. By extension, environmental compensation for hypobaric stress in plants is not as simple as increasing the oxygen content as was done for humans in the low pressure environments of the early space age. These data were published in the journal Plant Physiology (cover article) in January 2004 (Paul et al., 2004). In 2005 the analyses of the mature arabidopsis plants to evaluate the effects of exposing plants to hypobaric environments of increasing severity have been completed. Plants were exposed to atmospheres of 101 kPa (control – normal atmosphere at sea level), 75 kPa, 50 kPa, 25 kPa and 10 kPa for 24 hours then harvested for RNA extraction and analyses. RNA was prepared for all of these samples and they have been evaluated for global changes in gene expression patterns with Affymetrix microarray gene chip technologies (see Paul et al. 2004 for details). The data indicate that even very small changes in atmospheric pressures have a broad impact of the mobilization of the arabidopsis genome. The experiments initiated in the late spring of 2004 are still under analysis and builds on these findings. They are described briefly below, along with their respective status of analysis. Variable atmospheres and developmental stages – plate grown arabidopsis plants. These experiments were conducted at the University of Guelph in collaboration with Dr. Mike Dixon. This set of experiments was extensive, and designed to be comprehensive enough to address the first three aims, and thereby lead to the development of strategies for aim 4. The experiments were conducted throughout the month of May in Guelph and are comprised of five parts. Particulars of the experiments conducted at the University of Guelph were detailed in the 2004 Task Book. In 2005 the microarray analyses for many of these samples have been completed. The experiment conducted with 10 day old plants investigating the effects of variations in atmospheric pressure (nine different atmospheric pressures for a period of 24

	hours: 101 kPa, 75 kPa, 50 kPa, 25 kPa, 10 kPa, 5 kPa, Ox1, Ox2, Ox3) are essentially complete and are being developed for publication. The analyses required 62 microarrays: arabidopsis root and shoot samples from the 9 pressures above, multiplied by three as they were conducted in biological triplicate (three separate chambers for each pressure treatment) plus eight samples representing tissue culture callus cells at various representative pressures. The analyses of the microarray data have been completed and are in the initial stages of being organized for publication. Abcissic Acid (ABA) mutants and adaptability to hypobaric conditions. Genes associated with ABA metabolism were among the most prominent examples of differential gene expression in response to hypobaria. Arabidopsis lines that contain mutations in a variety of ABA related pathways are being examined for their ability to cope with hypobaric environments, and dissected for altered patterns of gene expression in response to hypobaria.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Paul A-L , Ferl RJ. "The biology of low atmospheric pressure-- implications for exploration mission design and advanced life support." Gravitational and Space Biology, in press, November 2005. , Nov-2005
Journal/Magazine covers	Paul A-L, Schuerger AC, Popp MP, Richards JT, Manak MS, Ferl RJ. "Cover in journal Plant Physiology for article, “Hypobaric biology: Arabidopsis gene expression at low atmospheric pressure." Plant Physiol. 2004 Jan;134(1):215-23. PMID: 14701916 , Jan-2004
Presentation	Paul, A-L.; Schuerger, A.; Ferl, R J. "The Future of Space Exploration in Canada Genomic Responses of Arabidopsis to Abiotic Stresses Relevant to Mars Exploration and Colonization – What do we need to know before we go? " The 5th Canadian Space Exploration Workshop (A-L. Paul invited speaker) May-2005
Presentation	Paul, A-L.; Schuerger, A.; Ferl, R J. "In Situ Biological Response: Scalable Assay Of Complex Biological Phenomena Using Genetically Engineered Plants" LEAG Conference On Lunar Exploration Oct-2005
Presentation	Paul, A-L; Schuerger, A.; Dixon, M.A.; Ferl, R J. "Low Atmospheric Pressure Plant Biology – Implications For Advanced Life Support In Non-Terrestrial Environments" American Society for Gravitational and Space Biology (R.J. Ferl invited symposium speaker) Nov-2005